

Day 7:Version Control Using Git

Introduction

Git is a distributed version control system used to track changes in source code and other files during software development. It was created by **Linus Torvalds** in 2005 to support the development of the Linux kernel. Git helps developers collaborate efficiently, maintain a history of changes, and manage different versions of a project. It is widely used with platforms such as GitHub, GitLab, and Bitbucket.

What is Version Control?

Version Control is a system that records changes made to files over time so that users can recall previous versions whenever needed. It allows multiple developers to work on the same project without overwriting each other's work. Version control improves collaboration, reduces errors, and makes software development more organized.

Features of Git

- Tracks changes made to files and source code.
- Supports collaboration among multiple developers.
- Maintains a complete history of project versions.
- Allows branching and merging of code.
- Enables rollback to previous versions if needed.
- Works offline because every developer has a complete copy of the repository.
- Fast and efficient for managing software projects.

Common Git Commands

- **git init** – Creates a new Git repository.
- **git clone** – Copies an existing repository to a local computer.
- **git status** – Displays the current status of files.
- **git add** – Adds files to the staging area.
- **git commit** – Saves changes permanently with a message.
- **git push** – Uploads local changes to a remote repository.
- **git pull** – Downloads and merges changes from a remote repository.
- **git branch** – Creates or lists branches.
- **git merge** – Combines changes from different branches.

Advantages of Git

- Free and open-source software.

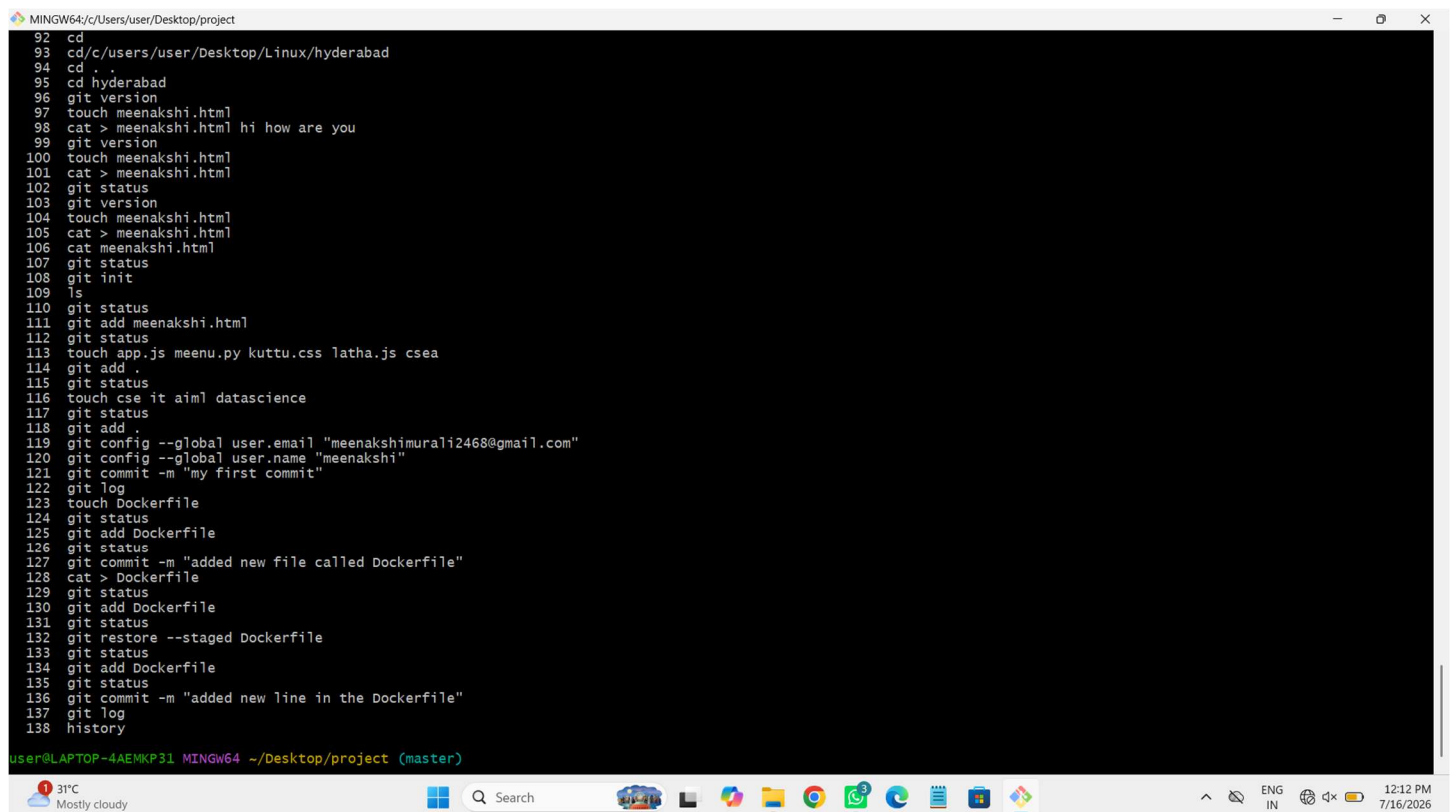
- Supports distributed development.
- Provides fast performance.
- Easy backup and recovery of projects.
- Maintains complete project history.
- Simplifies team collaboration.
- Widely supported by development tools and cloud platforms.

Applications of Git

Git is used for software development, web development, mobile application development, DevOps projects, open-source software, cloud computing projects, and collaborative team development. It helps developers manage code efficiently and maintain high-quality software.

Conclusion

Git is one of the most popular version control systems used in modern software development. It enables developers to track changes, collaborate effectively, manage multiple versions of a project, and recover previous versions whenever necessary. Its powerful features, speed, and flexibility make Git an essential tool for individual developers and software development teams.



```

MINGW64~/c:/Users/user/Desktop/project
92 cd
93 cd/c:/users/user/Desktop/Linux/hyderabad
94 cd .
95 cd hyderabad
96 git version
97 touch meenakshi.html
98 cat > meenakshi.html hi how are you
99 git version
100 touch meenakshi.html
101 cat > meenakshi.html
102 git status
103 git version
104 touch meenakshi.html
105 cat > meenakshi.html
106 cat meenakshi.html
107 git status
108 git init
109 ls
110 git status
111 git add meenakshi.html
112 git status
113 touch app.js meenu.py kuttu.css latha.js csea
114 git add .
115 git status
116 touch cse it aiml datascience
117 git status
118 git add
119 git config --global user.email "meenakshimurali2468@gmail.com"
120 git config --global user.name "meenakshi"
121 git commit -m "my first commit"
122 git log
123 touch Dockerfile
124 git status
125 git add Dockerfile
126 git status
127 git commit -m "added new file called Dockerfile"
128 cat > Dockerfile
129 git status
130 git add Dockerfile
131 git status
132 git restore --staged Dockerfile
133 git status
134 git add Dockerfile
135 git status
136 git commit -m "added new line in the Dockerfile"
137 git log
138 history

user@LAPTOP-4AEMKP31 MINGW64 ~/Desktop/project (master)
  
```